The Shift Toward Natural AI: Dr. John Henry Clippinger on Active Inference & the Future of AI

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[Music] hi and thank you for joining us again for another episode of the spatial web AI podcast today I have a very special guest John Henry cliping Jer is with us today he is the co-founder of biofor labs uh he is also the uh founding adviser for the active inference Institute and he is a research scientist for the MIT lab City Sciences groups uh John thank you so much for joining us today Welcome to our show well thank you for having me it's it great to be here U let me a little more context to my background as well see is it relevant to this conversation um yes I I've been at MIT the media lab um City Science group and before that with Sandy pentland and his human dynamics group uh and he and I developed a started a foundation called ID cubed um so sort of on sort of digital institutions decentralization we sort of anticipated blockchain things like that um much of about privacy and self- sovereignty and things like that and before that I was at Harvard at the um Berkman Center um and founded something called a law lab um and my background to get my PhD a long time ago in Ai and Linguistics and written chartism companies around that so I've had a long interest in artificial intelligence um in the early form and been tracking it um and so when it really sort of crossed this new threshold um that I became really re-engaged and felt like okay this is this is a whole new era this is It's really beginning as a science it's just not a lot of collections of of sort of programming techniques um and so became really really interested in it in particularly with the work of Carl friston I felt that I was I felt that his work really sort of pulled it together pushed it into a whole another threshold I've been involved I said Ser several companies and artificial intelligence particularly natural language processing wrote a book on that um but it would and been interested in to involved with the Santa Fe Institute and complexity science and self-organizing system all that um and that tied into the stuff of blockchain and autonomous decentralized autonomous organizations but it was really Carl's work that I think pulled it together tied it into physics gave it a principled way of thinking um and it's quite transformative for me um and actually that's how I I I first came to it through um I just discovered it went and looking at a a lecture that uh Maxwell ramstead did he really you know and I got to know Maxwell and what he was doing and really impressed and then we got on a number of the calls and networks that he had with Carl I see what's going on I worked with Maxwell a projects um but very very uh very u in in very sort of taken by what they were able to do um and then Maxwell went I actually introduced him to verus the people the spatial web Foundation I was very interested in spatial web Foundation early on because I think the whole idea of spatial Computing and beh a principle Bas of protecting data Ingy is really F foundational and we're moving into a new era where we're explaining things by models really I mean that and and became digital Twins and that again ties into Carl's work so all this sort of comes together um and I think we're moving in a whole new era um the motivation um for well partly just back up it's also a director of the active inference Foundation um and and in in working in in being that capacity really interested in saying how do you get people to understand it where it's going um and I've always been involved in open source software uh I've been very interested in that and and making a public good that's been part of my career and interest um and that's part of the rationale for the active inference foundation and then I was also part of something called the Boston Global for which is a group um it's in Boston but it's really Global um and there the interest was uh increasingly in artificial intelligence saying how do we create safe artificial intelligence how do make it a public good uh the interest spent not just in the US but around the world um and that form is an attempt to bring in people cross disciplines um there's a big disconnect as you might imagine between policy people and uh tech people i' I've been in government I done a lot of policy work uh so I know that they speak two different languages and and and I think this is so it's really important not to have them sort of the policy makers look at it an afterthought but really understand the principles and how transformant it is that was really so a little background and context for why uh I'm I'm engaged on these particularly writing the letter that we did yeah you know and um just to back up uh to what you had said about verses I actually was speaking with Dan uh the other day and I told him that I would be interviewing you and he said that they really owe you so much gratitude because you were you played a real significant part in their early days introducing them to Maxwell and therefore Carl as well so yeah well thank you you said that yeah no I was really interested in what they're doing they were really ahead of the trend um and U the fact is going through the i e I think is pretty profound um yeah and and um so I I saw the connection there um but I think what we're now where I am now thinking of working um is really and as part of the idea of biofor labs our collaborations with different people is that we're seeing the transition in AI it's going to be in everything um and you really do not have a a principal governance approach to it I was very much involved in privacy and this whole area of self-sovereign identity and data at at burkman Center World involved the world economic forum and working some of those policy I was very concerned that you really that regulation is is always much slower and and it's sort of like an afterthought and sudden they slap a lot of things on and you have a lot of adverse effects you really don't understand that people are playing to the trends and things like that um and so it's really as I said earlier it's really important to get a under principal understanding of these things and the current narrative is so much is so much captured by the the these extreme ex claims about the you know the AGI the superintelligence taking us over as arel Schwarzenegger The Terminator you know to the end times and and I and I I think that actually it's it's a very powerful technology that's an understanding INT in my view intelligence and life and what it means to be life are are really interconnected I mean Carl makes his point you know it's it it's like I alive because I I I can make predictions and I think Andy Clark his new book makes this point very very well um and and so what is information what is energy what is intelligence what does it mean to stay alive there there are many kinds of intelligence um and I think that's in the work that you find in Michael Lev that's a very distributed thing it's not what a kind of thing there's not a uniform monolithic intelligence I think um a lot of this is just projecting onto the machine now we're starting to see the limitations of large language models and Transformers um yeah and they start to be really apparent um I wrote a piece I calling it a a it could be called a a a a parlor trick like a Ouija board because people are pushing it around it's a Ouija board with memory and then they sort of ascribe certain things to it um and I I think yet I do think there's some really fundamental things that's very profound that's been done as is a I think they've did they they really the transform models very powerful in understanding semantics and things we didn't before I mean so it's really I I don't want to dismiss that at all but I do think the way the the global the current narrative is is is that okay it's only going to be running really big models you got to give it to Big Tech I don't think you have to run big models I think you need caal models I don't think you all those parameters blah blah blah all that so I think we're going moving to that new transition the next generation is coming about yeah you know and it's interesting that you say that because to me that's how I see this too is that you know I mean when you look at human knowledge right and and these you know uh autonomous systems are going to be extensions of human knowledge and the way human knowledge grows is by A diversity of a lot of different Minds a lot of different intelligences a lot of different levels of understanding and and viewpoints and all of that that kind of converge and interact and it makes this Global Knowledge get bigger you and you can't do that with one brain or one big machine you know you'll just have a reiteration of the same information there won't be growth to there won't be expanding to the knowledge from there um is that how you see it as well I totally agree I I I think that I mean you see this in nature I mean that just is there are many different kinds of body types that adapted to different situations there's not one there are different kinds of intelligence intelligence and body typ go together you know they're different kinds of life forms that exist and they have at different kind of niches and they and they op and then they combine together to create different ecosystems and so we're really understanding s of the biology of intellig is the biology of of of of what it means to be a living thing and I and that's very different than the mechanical reductionist notion um and that's why I think it's it's it's really important to uh keep this at open research we're just at the beginning of understanding this I think is is analogous to where synthetic biology was in the sense that you were experiments and this came out you really want to keep it open in growing and growing and and being able to involve lot of people in being able to develop a good tribute to those intelligence that that's that's been my posture that was sort of a motivation for the for the letter as well yeah and so I I really do want to get to the letter um and you know maybe I I I was going to start in a different spot but you know to continue with what you're talking about one of the things that was really interesting to me in the letter and I think um it was when it it was talking about these machine learning models the Transformers the llms and that uh engineering approachment and advancements made were made with no scientific principles or independent performance standards referred uh referenced or applied to the direct research and development and you know that's a that's a really important point and so I maybe we can take it from there and start there with why is that a problem well I I I see I think Ai and as I really involved it many many years I was at at the MIT AI way back when and and so it was really it was not there was not any underlying science there wasn't it was wasn't vers based on first principle it was really computer science being applied to a problem to see what we could do and you so you had many leaps and lurches around that you had symbolic Computing you had certain neural Nets and what you could do with neuronet and yes you had a molic and pits they had neuronet model isn't approximate to a physical model but then I think what you're now seeing and this is what again it gets back to Carl I mean I have to give it to him is going back to physics and first principles in physics then the hamiltonian the least action free energy you can actually start to build systems based upon first principles and testing against first principles as a science that's a huge shift I mean if we think we can have look at they said the physics of the mind and how the mind works and with the same rigor that we can start to understand the physics of physical system or even biological systems and make claims about that then that really changes how we look at our social institutions or Economic Institutions the whole idea of extended cognition I think is extremely P profound profound and you need again you need to base it on first principles so you know so we have been working with various people like uh not just Carl but also Michael Lin and other Chris fields and saying okay how do we lay down a foundation that's testable in a scientific sense that we can make certain claims yeah that and then so it and and and and then validate a very powerful notion is is this scale free domain free well is it can we really prove that can we really show that that the same principles work in different scales that if we can do that that really changes the game um and I think we can do that um so that that that's why that's so important well okay so then let's talk about the letter so uh you know you played a key role in uh writing this letter and you know what was the inspiration what was the purpose for it why did you think now is the time and you know what was the the goal in uh writing this letter I mean it really was was sort of I mean a frustration um that when you're hearing the public narrative captured to a certain point of view and and and that having been in the field for a really long time and talking to other people and saying wait a second this is this is Being misf Framed and that was it misf framed then that the misframing is being exploited by different parties to push a particular View and business model on top of this which I which I I don't Lily agree to so I I really felt and I and I sort of tested the waters and I said am I the only one and no it resonated with a lot of people um and so of the others that joined you in this well I mean so I mean one of the first was Chris Fields but then Carl came in and Carl was very sympathetic too of it um but then there was a whole other there were some of the people in the in the active INF world but it was broader than that was said it reached out to a broader Community but also through Boston Global form were able to reach with people who were um on the policy side um and and I really wanted to make that link I just didn't want to make it just a tech or science view but saying yes we we're trying to bring together different people to really think about this thing freshly and not have it be captured to a small particular interest spinning out of spinning out of Silicon Valley basically um and so um that that was really the motivation for it so we went through a number of iterations and letters with David silers Swig who was uh dean of the Harvard Medical School and also uh head of the um Neuroscience division at Harvard um he was very active in that he contributed to it because he's very interested in in developing a new way of thinking about mental health instead of using the these principles and looking at mental health beyond the skull was say Collective mental health can we what we see in can we link what's going on so is there a way in which we can see these things together so he he in Carl saon and that um so there's a bigger Community that's coming yeah I was just say what's really interesting about that particular um notion is I think a lot of the mental health issues that we're experiencing these days are a result of our interaction with tech right you know I I tot agree and we have no see I'm I'm a little bit fanatical about that because I do I mean there other there other because I I think that what we've done is is we've created algorithms of of exploitation and depression and addiction at a social level and we monetize that and we don't have a a policy means of saying hey that's that's a it's an opiate where're they're selling opiates out there yeah we have a principled way this say we do medicine say that is an opiate can we say that's an opiate and a cognitive opiate that's a cognitive opiate that's a cognitive addiction so I'm working on a paper on so the whole notion of uh when when ideologies become pathologies can you really say an ideology is a you know is a collection of basian beliefs that per form in a particular way but did they exhibit the patterns of something that's addictive or like even like cancer is it is it is so can we really apply those terms of racy to to metal States and Collective metal States I think so and but you can't you can't show it out there's a metaphor you really have to actually ground it in your science you know right so we're looking at being able to Carl started early work on this but David David's involved in this and being able to say okay we can do a a scan of the brain and map the the characteristics of that to on a collective at an individual level and see what's happening that changes policy big time yeah so it also gets into misstatements so is this a misstatement or not do we have a princip way of be saying what is what is hate speech what should what what isn't it what is the misuse of speech to create an addiction or some some other and can we base that on a principle way do we have a way of of stating rigorously what a fact is so I can go on and on uh well it's because there's a whole overlap between scientific method and and jurist Prudence in other words what is a fact how can I prove it in science and then in jur what is evidence what's a consensus what are the rules of evidence what kind of conclusions I can draw and can I have a computational model can I have something computationally represents that in a rigorous form that I can replicate it and explain it and you can't do that with LM bottles you can't do that close but you can do that with these new modles oh and what's interesting about that is that you know I mean we're we're entering into a future where we're going to have these intelligent agent assistants like that our life assistants our teachers our our mental health professionals well maybe we'll get a better sense of their mental health than some of the mental health professionals I mean you can actually you you can actually I mean what I'm interested in is is is there a grand truth that you can establish it's never an absolute truth but you can upgrade it and you can understand what it's based upon and so do I trust this agent do I how far do I trust this agent and this this again is where active infant is very good because it gives you a a variational free energy principle for measuring the level which you have uncertainty about certain claims or data and so you you and you could upgrade that so it's very overt about what it doesn't know and I think it sort of codifies the scientific method as I think can be a principal of governance that that's the other thing I'm interested in yeah so and and I'm sorry because I interrupted your train of thought on this letter and for the viewers who are AR aren't sure exactly what we're speaking about here with this letter it the the title of the letter was natural AI based on the science of computational physics and Neuroscience policy and societal significance right and um so please continue with with how you know you gathered all these people you realized everybody is on the same page sees kind of the same the same need and significance of uh really bringing this next generation of artificial intelligence to the public forum um yeah continue please well I mean I think I mean what what what has happened is that there now it's picked up it's another phase we're saying well how how do we encourage different groups to adopt this particular perspective so we have been approached by universities and others and other research groups to say yeah we we we we we we like the central proposition the natural a keeping it open and how can we develop repositories and things that actually function this way um and and so we're in in the early phases of that um but I I it it hit Accord with a number of folks um and I think it's I really do think you're you're seeing sort of the the exposure the the cracks in in in the current model um and then people exposing it and now they're have asking these questions and so where do we go from here and how do we shape that and how do we really make sure that uh it doesn't become captive and and it grows um as a public good as a public resource right world we are yeah and in the letter you know um you talk about pro-social benefits of natural intelligence uh approach so you know maybe you could speak about that um you know and and what are the pro-social benefits well I mean I think there there are many I mean I there there's several things that I we mentioned the whole idea of collective sort of collective mental health or extended cognition and being able way of framing that I mean the other area that I'm very interested in been involved in years um is in climate change um and uh so I've involved in a company is called clear trace the we we actually start with nfts that validate certification of production of of uh green energy solar primarily um and then opener Foundation was one of the founding directors of that now moving with working with some other projects and the Big Challenge and uh the Big Challenge to climate change is is being able to uh have people believe the measures you have the esgs you have all these other measurements and you have measurements that are in different domains of different scales how do you relate them um in is there an underlying organizing principle I I actually had talked to Carl we organized a call with Carl number years ago uh talking about something like digital guy cep that because he wrote a piece on on the on the uh Earth is a living system the planet is a living system the question and I think that is the right perspective and the question is how can you start with different kind of measures that reflect the uncertainty associated with and therefore that uncertainty is associated with an allocation of um a financial instrument to reflects that uncertainty but as the uncertainty gets less you reduce un certainly then you affect his financial instrument so we did I worked in some projects um with the the government of um Costa Rica on how to uh and basically they're interested in saying how they look they have 30% of of their their land masses sort of set aside as a natural reserves is there a way they can value that um as a natural asset so again you have assets that are based on nature now how do you how do you value that and as you get better and better measures how does that reflect the the value of the instrument or the underlying asset so this is where active inference comes in um and there's a new and so you you can tie your measurement uncertainty into the risk associated with a financial instrument so working with other people on that interesting um so that there's a group um number of groups are really interested that that's a big issue and if if you can crack that nut then you get a lot of people people to contribute they believe in the met this is why they believe in the metric because it's inexplainable and therefore they can assign that value to it and then you can attract Capital then you could scale it so I I think that is the only way we're going to really address climate change in scale um and so it comes back to really having a valid way of measuring things that's transparent that upgrades itself the reflect it what it doesn't know and that reflects itself in a particular kind of bond or financial instruments yeah you know it's interesting because that's one of the things that I I've recognized with the um you know with what the spatial web protocol will do in you know taking us into these 3D digital twin spaces that are now programmable because then once everything becomes a digital twin you can run simulations and you can't really argue with it because right now you have scientists that are like you know we see this as the potential outcome we see this as the potential outcome and then you have people just just going well I don't believe it you know I maybe it will maybe it won't but if you have concrete calculations that are telling you only I tell you I mean one of the things we experimented with was actually creating a digital twin of something and as your data gets better your resolution of the digital twin gets better so so so you you know you just think a tree was a very is high highly pixelated but then you get drilled down you understand more and more the Dynamics of the tree then you can predict more about this Behavior Therefore your your dat more about and you see that and people can work towards a common goal yeah so you you can have so this is like an active IMD you can submit policies around observations or actions you take in order to reduce the uncertainty and you can get compensated for that and so you have you have a principled way of getting people compensated that contribut to an outcome um and that that's to me is a new kind of governance principle and you can visually see it I mean so this is where it gets to the the the special web protocol they actually I can see something coming into being and I can work with other people and I get a very good model of it and therefore that that's that that's generated value and therefore there should be some kind of allocation of value for that yeah I mean you can really with a system like that you can create we spoke Futures you know you you know exactly what steps to take to get the outcomes that you're looking for because you're able to run it and and see the predictions huge and and and you can run many many use you can run explore a huge space that way so Autodesk has a lot you know the with auto desk I if you're familiar what they do but it's really worth looking into because you could you can say I want to design a c i can design a whole building and I can set certain parameters and then it'll generate different variants of it and then I can manage the consten see what I want and then I can send it off to a 3D printer and they can print it uh that's where we're going yeah yeah yeah fascinating wow okay so then you know you were talking too about one of the importance uh one of the important things of you know um introducing this this idea of the natural approach to you know these autonomous systems you know um and that you know it's going to really uh you know it's going to give us a lot of focus on all of the um potential outcomes of like AI as a uh opportunity for our future rather than a threat you know and that there are these you know existential threats that people are are talking about do you think that those really are potential outcomes bad outcomes if we stick with this you know direction that we're in with uh just focusing on the machine learning and the Deep learning I I I mean I the analogy I really I I I think um was it ad L was was wa minute she was one first she said well it it it's machines machines don't think is you you're projecting onto the machine what you think and Joe weisen ball did with the Liza program was like oh it was a very simple little program that simulated a shrink everyone projected everything upon it I think there's a lot of projection good and Ill so I I don't think it's necessarily in the in in in the because it has no intention it has no mind of so you're you're it's a it's a rejection and that's a rejection of the of the business model so people just displaying what what they're about I mean I do think that when I mean I did I did a certain amount of work with and and with government DOD with a whole area of network uh Network Warfare and then and decentralized command and control systems and so you look in in that world it's hard to tell I mean you're looking at what's happening in drones I and I know the bra has done interesting things and drones but you're going to have autonomous drones and and and and and the military is going to drive that and they can have so I it is a huge disruptor um and and so how these things get shaped is dependent by the institutional context in which they derived it's not it's not inherent it's the humans that will be stupid with the technology or they projector legacies now ideally our technolog is going to make us smarter you know I mean and and collectively smart it's not it it it's like we yes we're going to get these multiple intelligen we're going to know what we know what we don't know and we're going to be accountable to what we don't know and that's a big change we're enforcing accountability around own stupidity well that's good you know um and we don't have that so anyone can make any kind of claim and get away with it and you say wait a second no if we're g to have really valid model this is a protocol for doing that and we understand it we'll hold it to a high standard so that that I think is a great thing you know I think that's huge that's a very big plus um but you do have to be aware of this institutional context in which is developed um and that be and that's that's part part of my thinking is yeah creating institutional context that doesn't get twisted but actually takes the full value of it what are your thoughts on the misinformation side of it because with this I'm really big on that I mean I'm really big in trying to correct that and thing and I working on a piece and again it's like what is a fact what is evidence what can I Rely Upon and we do have method I the legal system absolutely depends upon that that's why you have a jury of peers that you rule on the evidence you have a whole set of processes to bring evidence so it but that you know that that is subject to all sorts of exploits and um and and so the but the whole point of of the sort of active INF modile is to actually is is a cify scientific method in such a way that you can actually update upgrade and be explicit about you're doing you're not saying you have the 100 the 100% answer no such thing as that but you can say I know this to a certain degree of confidence that I can upgrade it I can scale it it's totally transparent I I think you can eventually we're going to depend upon these kind of things for what we rely upon and trusted agents and certifications of fact I think that's going to be really really important so it's a Providence of the data where it come from and being able to replicate that to what extent I can rely upon it we have different tests and then I have agents that embody that and I'm willing to go with them and I can at any point I can expect them and ask them to explain themselves unwind themselves and go back and boret that becomes really critical again you can't do that you can't do that with the current llms no you know they're they're fed all kinds of information including a lot of inaccurate information and absolutely when they make it up I mean they just make it up yeah because it's so funny I I tell people this all the time you know um they're they're trying to you know spit out an appropriate output but appropriate and accurate are two different [Laughter] things they're designed to please whatever it's like a yes man yeah okay whatever you say poor lawyers are getting they've used it they legal I mean they did the review I think you saw that at Stanford or where it was I mean it's just it's just it generates nonsense and and and so you can read the witness very easy so you really have to curate the databases you really have to curate these things you have but once you do that and you have a trusted independent process is based upon first principle that's a real move ahead for us as a civilization that's a that's a real uper you know it's interesting because to me that's what that's the big advantage that I see with this approach especially when you consider like considering like what versus is doing and with the spatial web foundation and the spatial web protocol you know that's going to give these intelligent agents facts to develop their understanding of their environment their frame of reference it's going to be based on actual real time program data into all of the things and they're inter relationships and then they get real time sensory data too so they're actually dealing with real in the moment yeah I mean I mean so the sensor data and and and the you know the the whole internet of things the Zions of things that are out there I mean the one of the things so I was interested in blockchain and all that distrib governance and what you see with 5G and those whole networks the distributed system you get trillions of these things out there and there's something called son which a self-organizing network protocol is actually in use right now that these things correct themselves and and so they really have to if you can't trust those things then you're really in trouble right so they have to do it at the edge figure that at the edge and I and I think we we're we absolutely depend upon a trusted Network can verify itself that then allow the the better the greater inferencing to take place on that so how how do these systems cut through the noise of all the signals because there's going to be so many sensors and you know so um I know you know I I'd love to hear from you what you know you explain how the the self-optimization and the minimizing of the complexity I mean I you know if you could give your explanation of how this will function that I think our viewers would appreciate that um and you mean in terms of in well I mean I I think what you're doing is is you're you're creating models you're trying to identify CA of models that's the whole underlying process so you have you can techn you got a Quantum reference space you have something in multiple studies in hillbury space how do you be able to to project that into a three-dimensional space and then how do you able to infer out caal models and then how do you validate those CM models um and then I I think and then you're always updating them and so I do think that's this inherent in this whole process I I think the whole to to be intelligent is also able to compress complexity and and and be able to with with Main maintaining levels of fidelity right so that that that that that is that is the challenge um and then be able to know when you jump from one one state or one my beliefs into another state I mean because you may get locked in so what do you you is it exploit versus explore problem uh but there principal ways of doing that so I I I think um I I think that um you know I I I'm very positive about all that I I I think I share with you the optimism that goes along with it um it's just that we're in an Institutional context that that does not see the the economic value of that particular model yet and I I I think because they're still into the extractive mode um and um addictive algorithm versus generative algorithms of that's the what's crazy what's crazy about that to me though is that when you look at you know where the future of technology is going and the necessity for these um Smart City infrastructures and you know the possibility that we can build these smart cities that will become you know sustainable and inclusive and all these things that we need for our healthy population and a healthy Planet how could people not think this is like critical this is very much so this is part the the the in Ken Larson's group The City Science group this has been a lot of my involvement in that I mean they're building models of cities um and then they're building and you can change the models based upon the outcomes that you want and you can see the impacts of it and so it is they're trying to increasingly you know be able to virtualize a city and to then model different kinds of behaviors to see what kind of impacts you have on say carbon SE sration or carbon generation but also on pro-social Behavior so so that's one of the things that got me interested again implied in Carl's stuff is to model a community and say okay how does a community function how does it dysfunction if we be want to change it in order to make it more Equitable and open up opportunities what are the things that we that that what are what are the various things that affect those kind of outcomes and we can start to model that um and so I know Kent we're doing simulations of that Kent larsson's been very involved in that he's working with the a very famous architect norn Foster and they actually just had a big event last week so this this is the next Generation how to ma imagine the cities um but also how to make them really local make them enjoyable create the right kinds of human interactions and create the right kind of scale um and and give people agency with in in a sense of agency and significance within the context of a city make it really a living thing not a machine thing you know um I'm very partial to I've been work in Italy so I love the way Italians do think so you know it's a relation Arian economy you know and and and so people enjoy that moment enjoy the conversations of what they're doing how do you create that context allows that to happen um and so it this can definitely facilitate that how do you see that kind of interaction between human and machines in the future because we're you know we're going to have robots we're you know we're likely going to have robots in our homes we're going to have these you know virtual agents that are our assistants but have this uh embodiment to them this personification to them you know so how do you see that kind of you know Aral you know interaction of like well ier so I I I don't May these are extensions and manifestations of ourselves and artisanal economy is is is one of building relationships and valuing differences in people so you know I I so you can have a very eccentric robot is very and Italian friends what would they have the really odd hobbies and things they carry to Great extremes they may be the world's expert in a certain kind of one in a particular tarot well okay that's great that's a great conversation everyone knew i' to have that I mean I think the challenge is it with people in in it's a profound challenge is it feel that you're dealing with something that's that's that is not alien and is authentic and that you're not living in the you're living in a spoofed world that I think is going to be really tricky you know I mean that um and and because right now people try to spoof you for everything right I mean and so you know you fishing attacks and things like that um but what happens you know when you get get up you get a robot is modeling your mind and then trying to play to get your mind to see what I can do in order so you it's it's G to get it's going to get tricky I mean it you know it's gonna get creepy get really creepy right right I mean you could you know you could have a fact simil of yourself showing up and arguing with you and you a say and and wouldn't be that hard to have a replica of myself interrupt myself and telling me to get off the you know call right cloning with no genetics involved so I mean you know you could go some really strange places the other way but and so I it's going to be it's it's going to shake a lot of foundational habits and beliefs I mean there there's I mean yeah people are are terrified of what's going on right now I mean this you know it and I and and for kids and how to bring up kids and not have it be exploited and it's and just to throw something out there and and is oh you know oops sorry you know I I think that's not good enough um and and so yeah one you have to appreciate that so okay so if a uh an autonomous intelligence system agent can uh for lack of a better term clone your mind you know and really replicate you where does identity lie in that and proof of identity I don't know I mean I you know I I I was really interested because I was in a linguist and and Carl came out with this paper on communication and language and I think he's coming out another paper but in that paper I saw one one of the things that I got interest in AI about was like language was a singular human thing you know no one other creature has language and through language we can imagine other worlds we can simulate so it really is our our our special power I believe now lo and behold I was looking at Carl stuff I think you can have two agents come together and create synthetic languages this is if you think about in linguistics you have something know as a pigeon a Creole in that in a full language um and and so a pigeon is like two people never met each other they point to different things they establish certain kinds they ble to name things distinguish things establish certain predicates or relationships and then out of that dra a cre language then a full language I think what he has shown is is that there there are these sort of conventions are used to show they reflect in in the structure of the brain this is how we do this now we can do if we simulate that for agents and I can have a conversation with an an agent over a period to develop a private language for that agent um and and then and that can and I and and so that would be very very tuned to everything I do it's like a dog some a good dog watch everything you do you know a good a will watch and they'll figure out where you're going to go what you're going to do they'll figure it out um they'll do the same um and uh so that is that do that make you feel better or is that being watched I mean it's it's like it it depends upon context right I mean if if was a person and you really trusted that person you say I have a really intimate relationship if you didn't trust the person you think I'm being I'm being spied on I I'm being exploited right you just have a very opposite response right you know and you know it's interesting with what you say about language there because you know our closest friends and our closest relationships we do have these private bonds of communication you know nicknames for each other and you know inside jokes and you only have to say one or two words and the other person completes it that's right it's like it it it's it it brings you closer and it solidifies that Bond so you know uh to think of having these kinds of interactions with an autonomous agent if that autonomous agent doesn't have real emotion will it be the same will it just be felt by The Human Side of that kind of I don't know well I don't know I mean but you see I emotion I mean I think look we're talking about living things and we're looking at different thresholds I mean I I I I I don't think the emotions are off the table for for these autonomous living things I I don't I don't think they're really cognitive I think so I I think you have they have a they they have their own self-evidence and they self vested in being what they're going to be so they have fluctuations in their motion or their their cortisol levels or whatever about how what they're going to respond to and and and uh so you may have some of different personalities and you may get along with someone you may have a more intimate relationship with u the bot than the person I was I was on a it was really interesting because I was on a um I was judging a set of submissions in Vietnam for a bunch of high school kids um and and they were they were they were all special school and they were doing stuff in Ai and one of them had done a pretty exhaused study of of of um relationships uh you know romantic relationships with avatars and and uh yeah it's like it's a thing you know yeah oh I know there's like the replica app yeah yeah and so they were talking about it and how they build the relationships and different types of them had certain kind of feelings and kids and because the kids felt very alone and they and the Avatar talked to them and they really liked it it's like a it's like an imaginary friend for a kid you know it's something and and so it had real standing for it and it was really interesting um and you got a sense of a lot of isolation of those kids um and and so that forced another and and so there they're drawn to something that that's just recognized them was their own on the value them who they were um and so do we we treat people s surgus or do we treat people directly as people in order to do this I mean I I think it's it's it's surfacing a a lot of things um yeah but I can imagine uh mean people get attached to the car oh children babies to get attached to blankets and things blankets or and of course anal and they back anal and they and they really you know there's a real attachment there um so I I think it it's inevitable at some point I mean it really is yeah yeah no and and you know that that's that's going to be really interesting to see it unfold because you know like you said just in the things that we are witnessing right now you know that application replica you know where you have a a you know autonomous interaction with the spot you know what's really interesting about that is the first couple of years I guess um you know they they would let you upgrade to like you know kind of sexual uh interactions you know as far as dialogue and stuff like that and then at some point and I think it was early last year they cut that out because for whatever problems or whatever thinking that were arising and maybe it had to do with their uh their board of directors whoever that was saying no we can't really take it in that direction they cut it out and the people who had been in those engagements with their own Bots they really suffered uh mentally over it to the point where they they they went back to allowing it for the Legacy people who were already there because they there were people who were like suicidal they felt like they had had this relationship ripped away from them you know um this is really fascinating I mean it it is because it's what we think is real I mean is is is so that's real I mean right I mean to the each person involved it is a realaction it's real it's real something physical and we and and we we haven't quite made that leap and in into saying it's a reality there um and yeah no I I think that that that's going to be difficult um uh and I think they're going to be gener divides on this thing you know I mean in in cultural divides and stuff like that I mean when you think about how it disseminates and how rapidly it goes in the adoption patterns of what's allowed and what's not allowed yeah I mean I it'd be really tricky to be a parent and through battle this oh so agree with that I think I mean honestly I think we're seeing so much of that you know already you know and and even my kids my kids are all in their uh early 20s and um even when they were you know pre-teens and teenagers having to deal with the Social Media stuff and all of the craziness and the Damage Control that they had to be Su you know they were susceptible to every day you know you had these Anonymous there was one I remember when my daughter was in high school there was this one um app and I forget what it was called but it literally was Anonymous so anybody could make a statement on it about somebody and it started going through their school and it was tearing these kids apart because they were literally sitting there you know having to do damage control against all these rumors and all this stuff that was just you know going rampant very interesting I didn't I I wasn't aware of that I mean so I that was one of the things that I I was one of the first I was at har when Facebook was started and and I was one of the first people on Facebook and I took it this is and and took a a big objection to the fact that you have anonymity and you can't at certain points at that time they didn't protect kids and it was a big debate um but the idea of this sort of anonymous commentator being able to there's so much power it Taps into a darker side of people um and rewards that and and and and that whole adolescent venue I mean it it it's really destructive um and they're know anies they don't I mean it's it's you know what was fascinating to watch though was that there was Pro this this site came out right it was this storm for like three months or something and then all of a sudden it's like all of the kids they just like at the same time had this consensus of like forget it likeing it you know that's a very see I'm very interested in that because how that naturally forms I so can you do that can you what is what is the threshold that allow see this gets into my idea of uh pathologies and ideologies so how how did they develop a natural immunity to that I mean that's really I think they all just got so frustrated and they all real some something happened and they all realized they were fighting the same demon that was just didn't have like it wasn't there if they walked away and so they all just oh that's very interest that's a good way of phrasing it if they walk away it it disappeared that's really interesting yes that's why it it that's yes I mean because that's so many conspiracies so many fear-based things are constructed that way exploited that way um and now it's at the national scene I think oh that's really interesting yeah it was fascinating watching that because as a mom I'm sitting there going my baby like she and like know and to me one of the one of the CRA iCal things that I saw at that point was I needed to make sure that I was aware of what was happening like kind of parenting from inside the Huddle so that I knew what was going on because I knew that it was so necessary she would need my wisdom need my uh discernment like helping her guide her through this rather than just like you know being outside the Huddle and not what's going on and that's a delicate balance too that's real interesting so that if you could have an agent that could be a companion to kids and that they trust to provide the discernment and and allow that then you could help inoculate them from these kind of things oh I like that wow it's interesting yeah but yeah I do not envy any parents going forward they've got even more to deal with oh my gosh okay so so back to the letter back to that uh the you know the effort uh there in bringing this to public awareness where do you see that going from here what do you see as the next step that's really important to get this on more of a public stage and you know I me what I mean there a couple things that that we're interested in one is looking at first principles and this whole idea of establishing first principles they saying we're claiming that there is an underpinning of a science to this that that and and it's a testable science and therefore we can make certain claims and so we're we're trying to pull together different people to identify the standard those those first principles then also being able to have some kind of tests or experiments to validate that that's at the high level um I think at the at the other level is to educate people as to what this is about to help understand the differences between what we're talking a natural AI of intelligences uh and also across different disciplines um and I I think it's starting to it get hold with the the quote the AI Community I can see you know you you put up that the the debate between Jan and Carl and all that I think that was you and rubbish Carl he never says anything bad about do had a really force it out of so okay I'll call it rubbish well yeah I love the the introduction to that where you know the the moderator was like okay we've been far too civil for far too long and he was like you know I actually see us as you know partners in crime but we'll make this adversarial for entertainment purposes exactly you know we're so English to call something rubbish yesterday so true to form um but you know I mean that's a Davos and that certain elevation um that and and I I think that meta is is certainly headed in that directing rebranding and talking its own terms that way um and so uh all of this will get clear I I also think that you know running smaller love models and smaller platforms is going to be a big deal uh and and I I think uh and that that that again that could change the whole Ecology of this thing rather than having be centralized we talked about decentralized and be a have a whole a network of very validated nodes and things like that all that becomes really important um so it's it's really um incrementally trying to deal with people bring them together get a sort of uh Elevate the conversation across different kinds of groups that I think there's certain talking points you're very good at the talking points that how to how to how to explain this to people I mean you know when it started out I mean oh I I point people oh here's Carl's lecture listen to that one you do that to me I but he's really I like him I like everything doesn't it doesn't always car so and and so we're really at an early phase of of of how this thing is going to get reformed but it it will a year from now it will it will be it will be I think much more much more of that mainstream yeah you know one of the things that I one of the things that I've uh started uh this year I started a substack CH Channel because I wanted to have kind of a hub that could be a communication Hub but I I I cannot stand Discord for some reason it does not resonate with me so I was thinking substack because my articles can go there but we can also organize other things around it and one of the things that uh I'm starting in next week is one of the first ones is I figured every month I could put together some kind of an educational presentation or something around active inference and the spatial web and and you know just kind of where this technology is headed and so I I'm calling it The Learning Lab live and it'll be live streams you know with like a live Q&A chat whatever um but I figure that might be a really good way to just create some discourse around this make it to where you know people start to to gain more understanding but also bring some conversation you know uh I mean I I also think that that that when there is a reference a reference technology application out there that really embodies this and distinguishes itself then every is is Comm I mean it's commercially successful on that basis and everyone's eyes are going to go up um right so when people see the limitations of current language mod now actually here's one I can use and actually I can let it loose and I can count on it and and it's not going to do squirly things then that and then it actually helps me transform my business I think of my business in a very very different way than I've ever thought about it before and I think you starting to see those conversations come around so I you know in another year or so I think world will be in a different different place yeah okay so I I I know we're kind of running out of time here but um you know what if one last question I have for you what excites you most about the future of this natural path to Ai and particularly friston's work like where do you see it five years from now 10 years from now 10 years I can't imagine I I I that's like a whole another who knows I mean I think I mean to me uh this this has just pulled everything together so you know so I mean there's a number of different fields I think it's applicable to those different fields where it has the biggest impact for me is I you know I I think about this whole this social economic crisis we're and ecological crisis we're in I think I think for me this is a very powerful technique for dealing with a very a real existential challenge in terms of climate change and so I would I mean I I I think damn if it could gets applied there and we can really get together at scale and address these things and not be extractive but be generative to the climate and create but but we're creating more living things we have a whole way of organizing ourself this base upon life not the destruction life and that becomes our new organizing principle that going to change their economy that can be extremely positive so I think it I think that to me is is the best thing about it and I think that that it applied in that direction I think they will that's what I'm interested in and I and and and I I do think it it could work there but I I have no idea how get the right way but it I'm working at different pieces of it but some some way it'll coass and because it's inherent in how it's inherent how you have the best method of thinking about intelligence and what social organization economic organization based upon these principles that that's trans that's that's good you know yeah yeah I agree 100% oh yeah so John how can people reach out to you how can they uh where do they where do they find you on the internet you know how can they reach out to you if they have any questions or would like to know more well I mean I give my have two I give you two emails and I get to send that and I'm I'm on LinkedIn and then I I was answer loln I'm not a big figure on the internet I'm I I'm not uh I'm not uh at Dro at Droid at at uh being a influencer so I I so I I I don't I mean so you know you can reach out to I uh through those those venues that's fine and and I generally respond and stuff um but I can include those into the show notes so that people can out hey awesome and then um so John Henry clippinger thank you so much for joining us here today W it's great to have a chat with you and good luck and and I enjoyed it and certainly a thank you so much and thank you to everyone for joining us and we'll see you next time